

Anomaly Detection in Ship Engine Data

Contents

Problem Statement.....	1
Exploratory Data Analysis (EDA).....	2
Statistical Anomaly Detection (IQR Method)	3
Isolation Forest for Anomaly Detection	4
Evaluating 2D Visualisation Effectiveness	5
Conclusion	5

Problem Statement

This project focuses on detecting anomalies in a ship's engine to prevent failures and downtime. The dataset consists of six features monitoring engine performance. The objective is to develop a detection system that flags abnormal conditions before they lead to costly breakdowns or safety risks.

Features
Engine RPM
Lubrication Oil Pressure
Fuel Pressure
Coolant Pressure
Lubrication Oil Temperature
Coolant Temperature

Table 1 Features of the Dataset

A key requirement is ensuring that only 1-5% of data points are classified as anomalies to balance sensitivity and practicality. Both statistical and machine learning methods are applied to assess their effectiveness.

Accurately identifying anomalies will improve operational efficiency, reduce maintenance costs, and enhance safety. The findings will inform recommendations on the most suitable method for real-world deployment.

Exploratory Data Analysis (EDA)

Before applying anomaly detection techniques, Exploratory Data Analysis (EDA) was conducted to assess feature distributions and detect potential outliers. The dataset contained no missing or duplicate values, eliminating the need for imputation. Summary statistics revealed large variability in Engine RPM (61–2239) and an extreme Coolant Temperature peak (195.5°C), indicating possible anomalies.

Visualisation	Purpose	Key Findings
Boxplots	Detect outliers	Outliers observed in several features (see figure below)
Histograms	Show distributions	Some features exhibit skewness
Correlation Heatmap	Identify feature relationships	Weak correlations suggest PCA is unnecessary

Table 2 Visualisations used

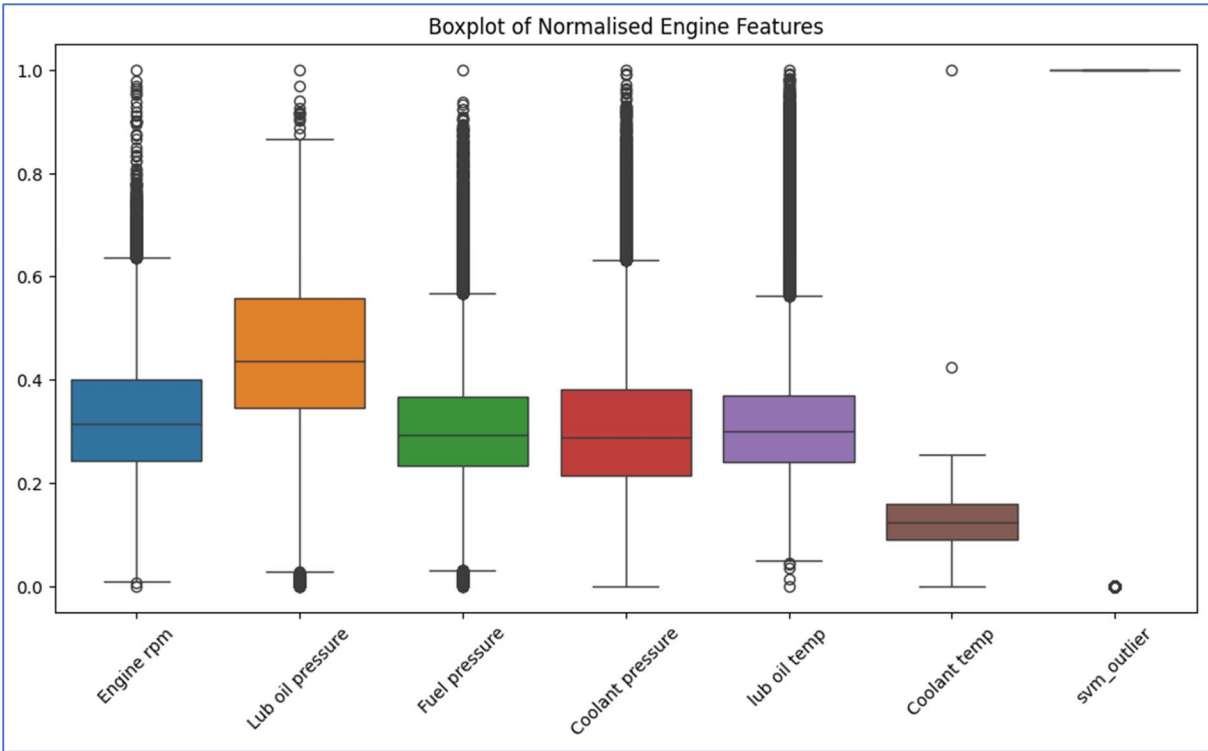


Figure 1 Boxplot of Normalised Engine Features

These insights supported an initial focus on univariate detection (IQR) to identify outliers per feature. However, as anomalies can span multiple features simultaneously, a multivariate approach was later explored using One-Class SVM and Isolation Forest.

Statistical Anomaly Detection (IQR Method)

The IQR method was applied to each feature to detect anomalies. While straightforward, its effectiveness varied. Some features over-detected anomalies due to high variability, while others under-detected due to tight distributions.

Feature	Outliers (%)	Within 1-5% Range?
Engine RPM	2.38%	Yes
Lubrication Oil Pressure	0.34%	No (Too Low)
Fuel Pressure	5.81%	No (Too High)
Coolant Pressure	4.02%	Yes
Lubrication Oil Temperature	13.40%	No (Too High)
Coolant Temperature	0.01%	No (Too Low)

Table 3 Individual feature outlier percentages

To prevent excessive false positives, a sample was classified as an anomaly only if at least two features were flagged as outliers. This reduced the anomaly detection rate to 2.16%, aligning with the expected 1-5% range.

Outlier Threshold ($\geq X$ Features)	Number of Samples	Percentage
≥ 1 outlier	4,636	23.73% (Too High)
≥ 2 outliers	422	2.16% (Within Range)
≥ 3 outliers	11	0.06% (Too Low)

Table 4 Samples with different numbers of outliers

While IQR provides a useful initial screening tool, it lacks adaptability for skewed distributions and complex patterns. To improve robustness, machine learning models—One-Class SVM and Isolation Forest—were implemented next.

One-Class SVM for Anomaly Detection

One-Class SVM detects anomalies by distinguishing normal and outlier data points using a hyperplane. Tuning parameters was essential to ensure anomalies were detected within the expected 1-5% range. The results are summarised below.

Gamma	Nu	Anomaly %	Within 1-5% Range?
0.1	0.01	0.97%	No (Too Low)
1	0.08	10.87%	No (Too High)
0.01	0.03	3.00%	Yes (Final Choice)

Table 5 Parameter tuning for One-Class SVM

The final model ($\gamma = 0.1$, $\nu = 0.03$) was selected as it identified 3.00% anomalies, aligning with the expected range. However, One-Class SVM was highly sensitive to hyperparameter tuning, making it more complex to fine-tune compared to other approaches.

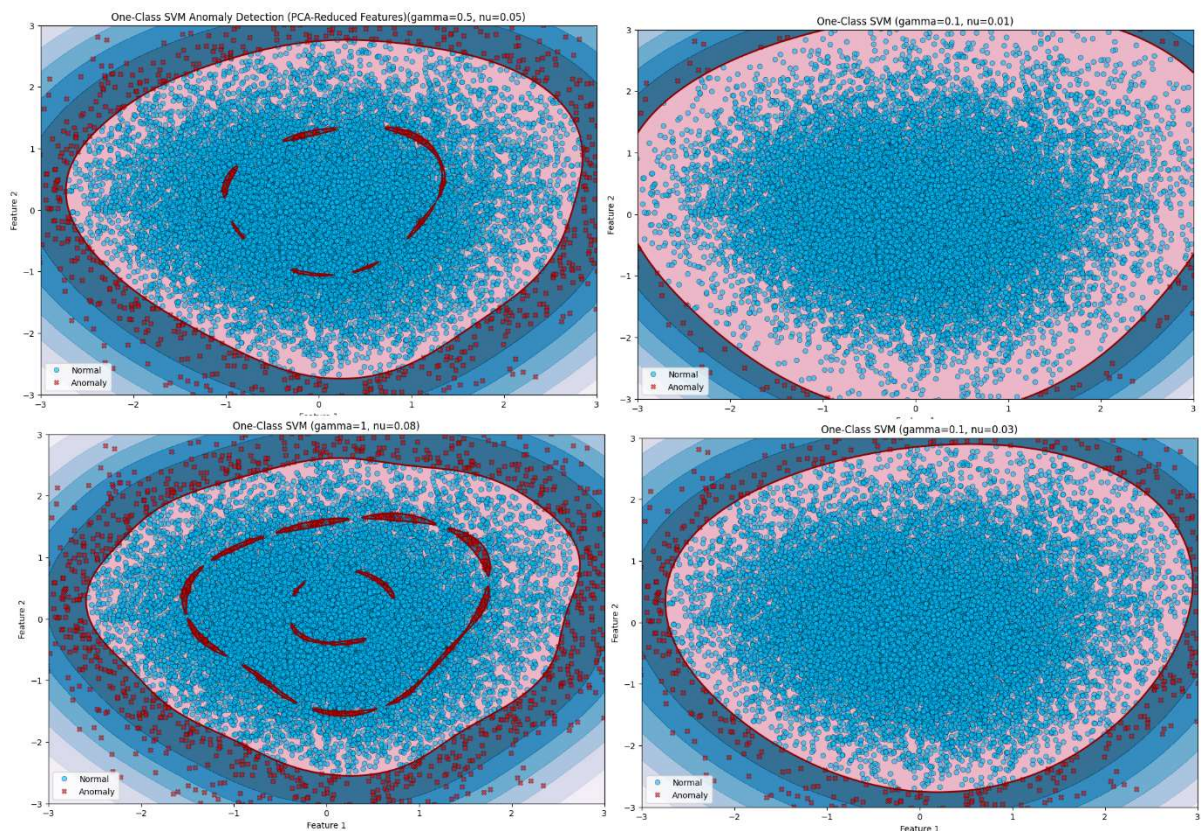


Figure 2 One-Class SVM Anomaly Detection (PCA-Reduced) for four different parameter combinations

Isolation Forest for Anomaly Detection

Isolation Forest detects anomalies by randomly partitioning the feature space. The key parameter, contamination, determines the percentage of anomalies detected. Unlike One-Class SVM, it is computationally efficient and allows direct control over the anomaly rate, making it the preferred model.

Contamination	Anomaly %	Within 1-5% Range?
0.01	1.00%	Yes
0.03	3.00%	Yes (Final Choice)
0.05	5.00%	Yes

Table 6 Parameter tuning for Isolation Forest

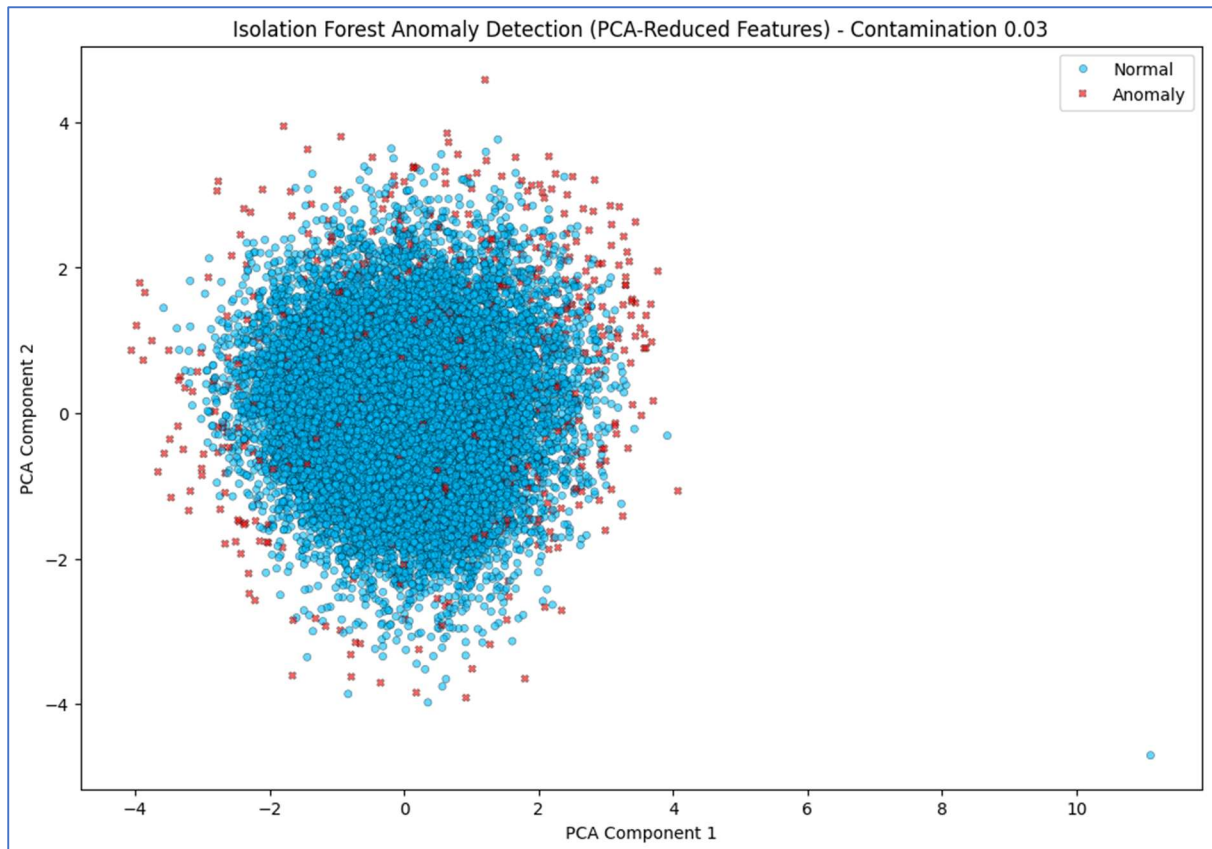


Figure 3 Isolation Forest 2D scatter plot using PCA

Evaluating 2D Visualisation Effectiveness

PCA reduced six features to two for visualisation, improving interpretability but introducing limitations. In both One-Class SVM and Isolation Forest, some detected anomalies were not visually distinct, while others appeared extreme but were not flagged.

For example, an outlier at PCA Component 1 ≈ 11 seemed anomalous but was not flagged by Isolation Forest. This highlights that PCA projections do not always align with model decisions. Anomalies must always be validated in the original feature space rather than relying solely on visualisations.

Conclusion

The IQR method provided a simple statistical approach but was inconsistent across features, making it unsuitable for general anomaly detection. Machine learning models offered better control, with Isolation Forest emerging as the preferred choice due to its efficiency and ease of parameter tuning.

Unlike supervised learning, where models are evaluated using accuracy metrics, anomaly detection in an unsupervised setting lacks predefined labels. Model selection is based on interpretability, computational efficiency, and control over the anomaly detection rate rather than absolute correctness.

Isolation Forest was preferred because its contamination parameter directly controlled the anomaly percentage, making it more practical than One-Class SVM, which required fine-tuning gamma and nu. Additionally, Isolation Forest detected anomalies efficiently without assuming feature distributions, making it more adaptable to real-world applications.

Method	Final Parameters	Anomaly %	Preferred?
IQR	N/A	Varies (0.01% - 13.4%)	No (Inconsistent)
One-Class SVM	gamma=0.01, nu=0.03	3.00%	No (More tuning required)
Isolation Forest	Contamination=0.03	3.00%	Yes (Best choice)

Table 7 Final Model Comparison

PCA visualisation helped in understanding anomaly patterns but had limitations, as seen in unflagged extreme points. This highlights the importance of validating anomalies in the original feature space rather than relying solely on 2D projections.

A hybrid approach combining statistical insights (IQR) and ML models (Isolation Forest) provides the most effective strategy for anomaly detection in ship engine monitoring.